Materiały dydaktyczne przeznaczone wyłącznie dla studentów wydziału Informatyki i Telekomunikacji

Przesyłanie danych multimedialnych

MPEG-TS - Transport stream

Packetized Elementary Stream

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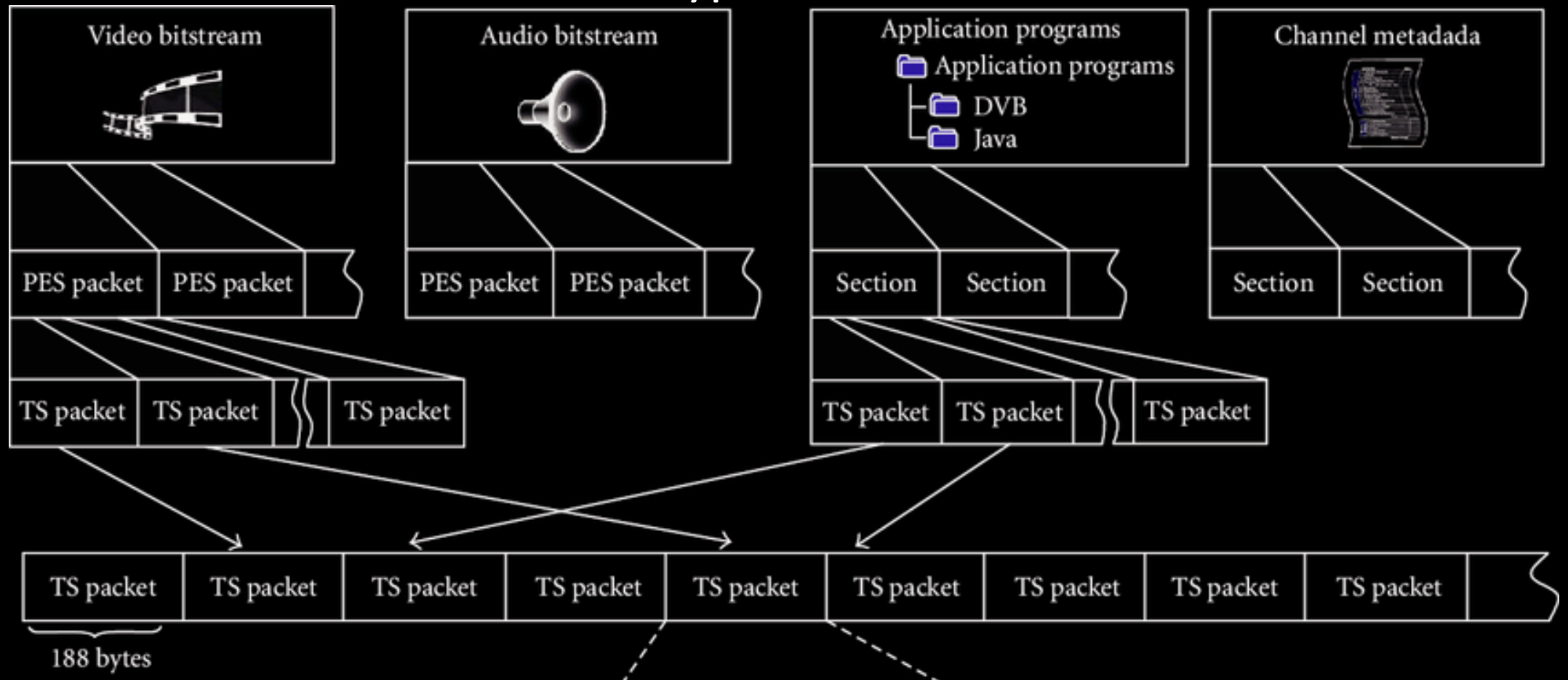
Przypomnienie



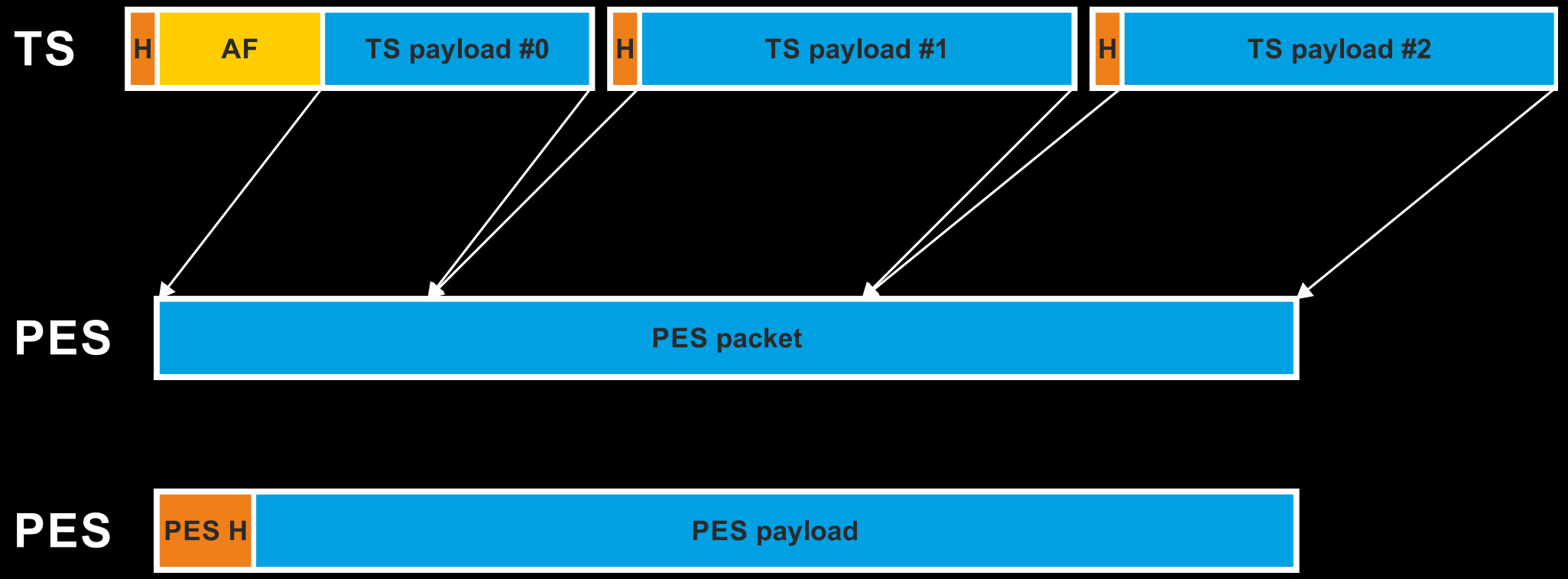
Field	Sync. byte	Transport error indicator	Payload unit start indicator	Transport priority	Packet ID (PID)	Transport scrambling control	Adaptation field control	Continuity count
Number of bits	8	1	1	1	13	2	2	4

Przykład

```
0000000000 TS: SB=71 E=0 S=1 P=0 PID=  17 TSC=0 AF=1 CC= 0
0000000001 TS: SB=71 E=0 S=1 P=0 PID=   0 TSC=0 AF=1 CC= 0
0000000002 TS: SB=71 E=0 S=1 P=0 PID=4096 TSC=0 AF=1 CC= 0
0000000003 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0 AF: L=  1 DC=0 RA=1 SP=0 PR=0 OR=0 SP=0 TP=0 EX=0
0000000004 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 1
0000000005 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 2
0000000006 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 3
0000000007 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 4
0000000008 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 5
0000000009 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 6
0000000010 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 7
0000000011 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 8
0000000012 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 9
0000000013 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=10
0000000014 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=11
0000000015 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=12
0000000016 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=13
0000000017 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=14
0000000018 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=3 CC=15 AF: L= 47 DC=0 RA=0 SP=0 PR=0 OR=0 SP=0 TP=0 EX=0
0000000019 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0 AF: L=  1 DC=0 RA=1 SP=0 PR=0 OR=0 SP=0 TP=0 EX=0
0000000020 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 1
0000000021 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 2
0000000022 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 3
0000000023 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 4
0000000024 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 5
0000000025 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 6
0000000026 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 7
0000000027 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 8
0000000028 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 9
0000000029 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=10
0000000030 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=11
0000000031 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=12
0000000032 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=13
0000000033 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=14
```

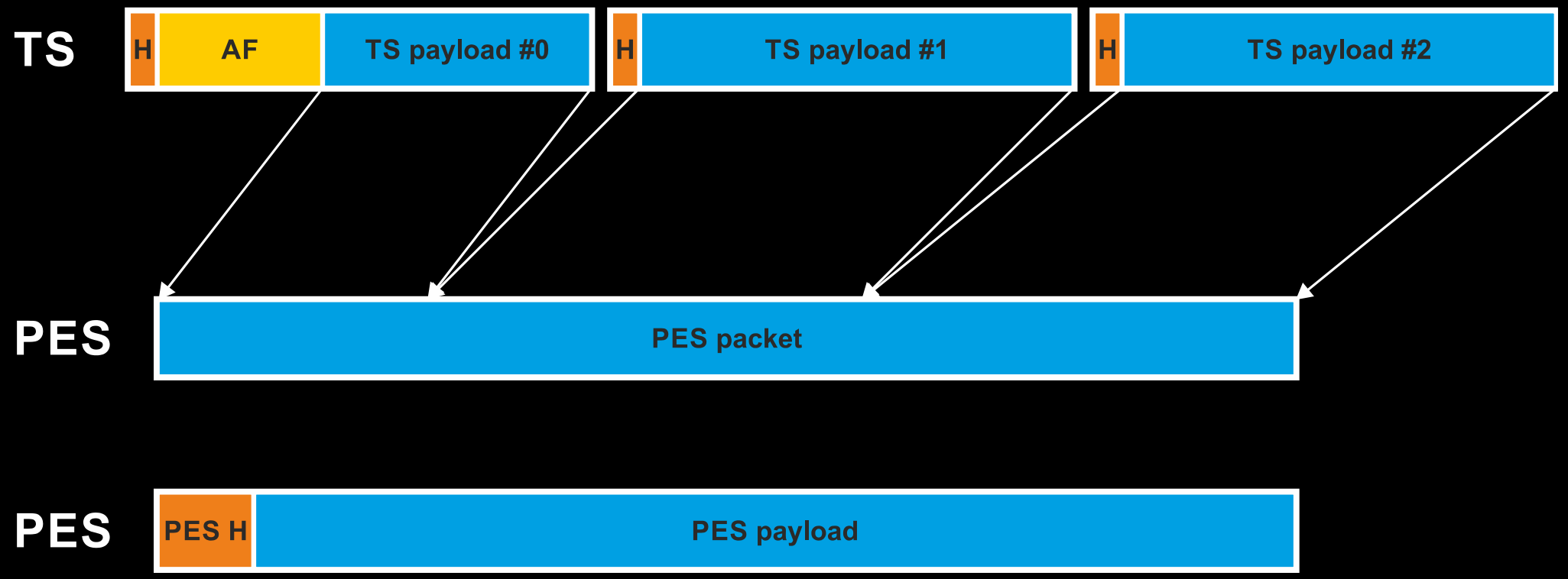
Packetized Elementary Stream

- Strumień elementarny – pakiet
  - header (nagłówek)
    - zmienna długość nagłówka (>=6 B)
  - payload (pakiet danych, np. ramka fonii)
  - ITU-T H.222.0 | ISO/IEC 13818-1
    - Section 2.4.3.6 – PES packet



Packetized Elementary Stream

- Strumień elementarny – nagłówek
  - podstawowe pola (pierwsze 6B)
    - `packet_start_code_prefix (24) = 0x000001`
    - `stream_id ( 8)`
    - `PES_packet_length (16)`
  - `PES_packet_length` może przyjmować wartość 0



Packetized Elementary Stream - Header

Table 2-21 – PES packet

Syntax	No. of bits	Mnemonic
PES_packet() {		
packet_start_code_prefix	**24**	**bslbf**
stream_id	**8**	**uimsbf**
PES_packet_length	**16**	**uimsbf**
if (stream_id != program_stream_map		
&& stream_id != padding_stream		
&& stream_id != private_stream_2		
&& stream_id != ECM		
&& stream_id != EMM		
&& stream_id != program_stream_directory		
&& stream_id != DSMCC_stream		
&& stream_id != ITU-T Rec. H.222.1 type E stream) {		
'10'	**2**	**bslbf**
PES_scrambling_control	**2**	**bslbf**
PES_priority	**1**	**bslbf**
data_alignment_indicator	**1**	**bslbf**
copyright	**1**	**bslbf**
original_or_copy	**1**	**bslbf**
PTS_DTS_flags	**2**	**bslbf**
ESCR_flag	**1**	**bslbf**
ES_rate_flag	**1**	**bslbf**
DSM_trick_mode_flag	**1**	**bslbf**
additional_copy_info_flag	**1**	**bslbf**
PES_CRC_flag	**1**	**bslbf**
PES_extension_flag	**1**	**bslbf**
PES_header_data_length	**8**	**uimsbf**

Parser strumienia transportowego

- Krok 3:
  - pobranie i otwarcie dokumentu ITU-T H.222.0 | ISO/IEC 13818-1
  - „składanie” pakietu strumienia elementarnego
    - dla wybranej wartości PID=136 (fonia)
    - parsowanie nagłówka pakietu PES
    - wyliczenie długości nagłówka
    - weryfikacja poprawności
      - prawidłowy PID, ciągłość CC
      - porównanie:
        - » wartości pola `PES_packet_length`
        - » wyliczonej długości nagłówka pakietu PES (włączając opcjonalne pole `PES_header_data_length`
        - » długości pakietu

- Zadanie dodatkowe 2:
  - parsowanie zawartości pól PTS i DTS

Przykład

Adaptation field

```
0000000003 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0 AF: L=  1 DC=0 RA=1 SP=0 PR=0 OR=0 SP=0 TP=0 EX=0 Stuffing=0
0000000018 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=3 CC=15 AF: L= 47 DC=0 RA=0 SP=0 PR=0 OR=0 SP=0 TP=0 EX=0 Stuffing=46
0000000019 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0 AF: L=  1 DC=0 RA=1 SP=0 PR=0 OR=0 SP=0 TP=0 EX=0 Stuffing=0
```

Packetized Elementary Stream

```
0000000003 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0 Started  PES: PSCP=1 SID=192 L=2888
0000000004 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 1 Continue
0000000005 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 2 Continue
0000000006 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 3 Continue
0000000007 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 4 Continue
0000000008 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 5 Continue
0000000009 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 6 Continue
0000000010 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 7 Continue
0000000011 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 8 Continue
0000000012 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC= 9 Continue
0000000013 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=10 Continue
0000000014 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=11 Continue
0000000015 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=12 Continue
0000000016 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=13 Continue
0000000017 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=1 CC=14 Continue
0000000018 TS: SB=71 E=0 S=0 P=0 PID= 136 TSC=0 AF=3 CC=15 Finished PES: Len=2894
0000000019 TS: SB=71 E=0 S=1 P=0 PID= 136 TSC=0 AF=3 CC= 0 Started  PES: PSCP=1 SID=192 L=2888
```

Szablon klasy xPES_PacketHeader

```
class xPES_PacketHeader
{
public:
  enum eStreamId : uint8_t
  {
    eStreamId_program_stream_map       = 0xBC,
    eStreamId_padding_stream           = 0xBE,
    eStreamId_private_stream_2         = 0xBF,
    eStreamId_ECM                      = 0xF0,
    eStreamId_EMM                      = 0xF1,
    eStreamId_program_stream_directory = 0xFF,
    eStreamId_DSMCC_stream             = 0xF2,
    eStreamId_ITUT_H222_1_type_E       = 0xF8,
  };

protected:
  //PES packet header
  uint32_t m_PacketStartCodePrefix;
  uint8_t  m_StreamId;
  uint16_t m_PacketLength;

public:
  void     Reset();
  int32_t  Parse(const uint8_t* Input);
  void     Print() const;

public:
  //PES packet header
  uint32_t getPacketStartCodePrefix() const { return m_PacketStartCodePrefix; }
  uint8_t  getStreamId             () const { return m_StreamId; }
  uint16_t getPacketLength         () const { return m_PacketLength; }
};
```

Szablon klasy xPES_Assembler

```
class xPES_Assembler
{
public:
  enum class eResult : int32_t
  {
    UnexpectedPID     = 1,
    StreamPackedLost  ,
    AssemblingStarted ,
    AssemblingContinue,
    AssemblingFinished,
  };

protected:
  //setup
  int32_t  m_PID;
  //buffer
  uint8_t* m_Buffer;
  uint32_t m_BufferSize;
  uint32_t m_DataOffset;
  //operation
  int8_t            m_LastContinuityCounter;
  bool              m_Started;
  xPES_PacketHeader m_PESH;

public:
  xPES_Assembler ();
  ~xPES_Assembler();

  void    Init        (int32_t PID);
  eResult AbsorbPacket(const uint8_t* TransportStreamPacket, const xTS_PacketHeader* PacketHeader, const xTS_AdaptationField* AdaptationField);

  void     PrintPESH        () const { m_PESH.Print(); }
  uint8_t* getPacket        ()       { return m_Buffer; }
  int32_t  getNumPacketBytes() const { return m_DataOffset; }

protected:
  void xBufferReset ();
  void xBufferAppend(const uint8_t* Data, int32_t Size);
};
```

Przykładowy kod w funkcji `main`

```
size_t NumRead = fread(TS_PacketBuffer, 1, xTS::TS_PacketLength, TransportStreamFile);
if(NumRead != xTS::TS_PacketLength) { break; }

TS_PacketHeader.Reset();
TS_PacketHeader.Parse(TS_PacketBuffer);

TS_PacketAdaptationField.Reset();
if(TS_PacketHeader.getSyncByte() == 'G' && TS_PacketHeader.getPID() == 136)
{
  if(TS_PacketHeader.hasAdaptationField())
  {
    TS_PacketAdaptationField.Parse(TS_PacketBuffer, TS_PacketHeader.getAdaptationFieldControl());
  }

  printf("%010d ", TS_PacketId);
  TS_PacketHeader.Print();
  if(TS_PacketHeader.hasAdaptationField()) { TS_PacketAdaptationField.Print(); }

  xPES_Assembler::eResult Result = PES_Assembler.AbsorbPacket(TS_PacketBuffer, &TS_PacketHeader, &TS_PacketAdaptationField);
  switch(Result)
  {
    case xPES_Assembler::eResult::StreamPackedLost  : printf("PcktLost "); break;
    case xPES_Assembler::eResult::AssemblingStarted : printf("Started  "); PES_Assembler.PrintPESH(); break;
    case xPES_Assembler::eResult::AssemblingContinue: printf("Continue "); break;
    case xPES_Assembler::eResult::AssemblingFinished: printf("Finished "); printf("PES: Len=%d", PES_Assembler.getNumPacketBytes()); break;
    default: break;
  }
  printf("\n");
}
```